

MULTI-AGENT URBAN TRAFFIC SIGNAL OPTIMIZATION IN SIMULATED ENVIRONMENTS

Project Id: 25-26J-523

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

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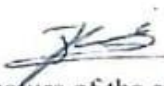
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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research focuses on developing a Model Predictive Control (MPC)-based traffic signal optimization module as part of a comprehensive multi-agent traffic management system. The module addresses critical limitations in conventional fixed-time and actuated signal control, which lack predictive foresight and adaptability to fluctuating traffic conditions. By integrating real-time data from SUMO simulations via the TraCI API, the MPC framework continuously forecasts short-term traffic and computes optimal green and red signal timings. The purpose is to enhance multi-agent traffic signal optimization by enabling constraint-aware, proactive decision-making that minimizes congestion metrics including average vehicle delay, queue length, and number of stops. The system employs advanced mathematical programming approaches and solves through optimization libraries, while incorporating safety constraints such as minimum green times. To ensure robustness, fallback mechanisms allow dynamic switching to rule-based controllers under computational delays or prediction errors. This research builds on recent advancements in predictive traffic control, optimization-driven intelligent transport systems, and hybrid MPC frameworks, contributing an explainable, scalable, and simulation-validated solution for adaptive signal control. The module ultimately supports the broader objective of developing real-time and operator-interactive traffic management systems for smart urban mobility.

Keywords - Model Predictive Control, traffic signal optimization, predictive control, rolling horizon, SUMO, TraCI, multi-agent traffic systems, real-time optimization, congestion reduction.

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LIST OF ABBRIVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BPNN	Back-Propagation Neural Network
CSV	Comma-Separated Values
CVXPY	Convex Optimization Modeling Framework for Python
DRL	Deep Reinforcement Learning
GCN	Graph Convolutional Network
GNN	Graph Neural Network
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
MAPE	Mean Absolute Percentage Error
MIQP	Mixed-Integer Quadratic Programming
MILP	Mixed-Integer Linear Programming
MPC	Model Predictive Control
QP	Quadratic Programming
REST	Representational State Transfer
RESTful	Representational State Transfer-compliant
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
UI	User Interface
VISSIM	Verkehr In Stadten - (Traffic in Cities - Simulation Model)
XML	eXtensible Markup Language

1. INTRODUCTION

Urban traffic congestion poses a significant challenge in contemporary cities, leading to extended travel times, increased fuel usage, and heightened environmental pollution [5]. Conventional traffic signal control (TSC) systems, such as fixed-time or basic actuated mechanisms, often rely on predetermined cycles that fail to respond effectively to real-time variations in traffic flow [3], [2]. These rigid approaches are inadequate for managing unpredictable events like surges in demand, incidents, or variable vehicle arrivals, resulting in prolonged queues, unbalanced intersection loads, and overall network inefficiencies [7]. Furthermore, while some systems incorporate rudimentary priorities for emergencies, they typically overlook predictive foresight and holistic optimization, exacerbating delays in dynamic urban environments [1].

To mitigate these deficiencies, advanced predictive control strategies have gained prominence, leveraging mathematical modeling and optimization for proactive TSC. Model Predictive Control (MPC) stands out as a robust framework, originating from process industries but increasingly adapted for transportation due to its ability to forecast system states, handle constraints, and optimize multivariable objectives over a receding horizon [5]. In TSC applications, MPC utilizes discrete-time models to predict traffic evolution—such as queue lengths and vehicle densities—based on inputs like green phase durations, enabling dynamic adjustments that minimize delays and balance flows [3]. For instance, early implementations focused on isolated intersections, employing state-space dynamics to estimate queues and optimize green times for reduced congestion [7]. More recent advancements integrate nonlinear queue equations and mixed-integer formulations to enhance accuracy, balancing queues through quadratic objectives rather than simple summation [2].

Hybrid extensions further amplify MPC's capabilities by fusing it with neural networks for improved prediction and decision-making. Approaches combining MPC with graph convolutional networks (GCNs) and deep reinforcement learning (DRL) introduce dynamic correlation matrices to capture spatiotemporal dependencies, allowing agents to explore optimal strategies under constraints [4]. Such integrations have demonstrated substantial queue reductions—up to 17 vehicles in high-traffic scenarios—and speed enhancements of 19.7% over inductive controls [4]. Additionally, MPC-based systems incorporating Back-Propagation Neural Networks (BPNN) for flow forecasting achieve low prediction errors (MAPE <12%) and waiting rate reductions of 29.7% compared to fixed-time methods [1]. A comprehensive survey underscores MPC's evolution over two decades, highlighting trends in coordination for networked intersections, integration with autonomous vehicles, and decomposition techniques for scalability [5]. However, persistent gaps include computational intensity for real-time deployment, reliance on accurate models amid uncertainties, and limited explainability in hybrid setups [5], [4],[10].

In essence, while existing MPC methodologies provide powerful predictive tools and hybrid enhancements, they often operate in isolation without seamless closed-loop integration or user-friendly interfaces for tuning and fallback. This project addresses these voids within the broader intelligent traffic management system. It develops an MPC-based predictive traffic signal optimization module that fetches real-time states (e.g., vehicle counts, queue lengths) from the SUMO simulator via TraCI API, forecasts traffic over a 30–60 second horizon, and solves optimizations (using QP for linear cases, MILP for discrete phases, or nonlinear MPC for realistic dynamics) to minimize delays, queues, and stops [5], [3]. Optimized signal plans are applied back to SUMO in a feedback loop, with novelties including constraint-aware handling (e.g., minimum greens), fallback to rule-based or GNN modes, and an interactive dashboard (built with Flask/Dash) for parameter adjustments [4]. This ensures explainable, proactive control, complementing the system's

simulation setup, AI-driven adaptation, and emergency handling, while overcoming scalability challenges through hybrid decomposition.

1.1 Background and Literature Review

Traffic congestion in urban areas has become a critical issue, driven by rapid urbanization, population growth, and increasing vehicle densities. Globally, it results in significant economic losses, estimated at over \$1 trillion annually due to wasted time, fuel inefficiency, and environmental impacts such as elevated CO₂ emissions [5]. In developing regions like South Asia, including Sri Lanka, intersections in cities such as Colombo often serve as bottlenecks, where inefficient signal timing leads to prolonged queues, increased accident risks, and suboptimal throughput [3], [8]. Traditional traffic signal control (TSC) methods, including fixed-time cycles and sensor-actuated systems, operate on static or reactive principles, failing to anticipate dynamic changes like varying arrival rates, incidents, or multi-modal interactions (e.g., pedestrians and public transport) [1], [5], [11]. These limitations underscore the need for intelligent, adaptive systems that can predict and optimize traffic flows in real-time, particularly through accurate modeling of traffic dynamics to forecast short-term states and support rolling horizon predictions.

Model Predictive Control (MPC) emerges as a promising paradigm for addressing these challenges, with foundational work emphasizing traffic dynamics modeling and prediction using macroscopic approaches like the Cell Transmission Model or data-driven neural enhancements [5], [10]. Rooted in optimal control theory, MPC uses a dynamic model of the system to forecast future states over a finite horizon, computes an optimal sequence of control actions by minimizing a cost function subject to constraints, and applies only the first action before re-optimizing in a receding manner [5], [7]. In TSC, the "system" is typically modeled as a discrete-time state-space representation, where states include queue lengths or vehicle densities per approach, inputs are green phase

durations or phase sequences, and disturbances account for stochastic arrivals [3], [10]. The cost function often penalizes metrics like total delay, queue imbalance, or emissions, while constraints ensure safety (e.g., minimum green times, inter-green intervals) and feasibility (e.g., cycle length limits) [2], [11]. MPC's application in TSC dates back to the early 2000s, evolving from linear approximations for isolated intersections to sophisticated hybrids for networked systems [5], [12], with predictive modeling enabling proactive congestion anticipation through hybrid models adaptable to stochastic arrivals [1], [6].

MPC's strengths lie in handling multivariable interactions, incorporating forecasts, and accommodating uncertainties through feedback updates, particularly in optimization framework development where solvers like QP or MILP minimize multi-objective costs while enforcing constraints [2], [3], [6]. For instance, by integrating real-time data from sensors or simulations (e.g., via SUMO's TraCI API), MPC can create a closed-loop controller that proactively adjusts signals to prevent congestion buildup [4], [9]. This is particularly relevant in integrated systems like the one proposed here, where MPC complements simulation setups, AI-driven adaptations, and priority handling to form a holistic intelligent traffic management framework [8], [13]. However, challenges such as computational complexity for large-scale networks and sensitivity to model inaccuracies persist, motivating ongoing research into efficient solvers and hybrid integrations [5], [6], [12], with advanced techniques like distributed or hierarchical MPC for network-scale coordination [13].

The literature on MPC for TSC is extensive, with key contributions focusing on real-time integration and API development for seamless data exchange and low-latency updates [4], [9]. A foundational survey provides a comprehensive overview of MPC methods in urban and freeway networks over the past two decades, emphasizing advantages in managing complex dynamics and evaluation metrics like delay reduction [5]. Solution approaches range from MILP for discrete phase decisions to nonlinear MPC for realistic saturations, with emerging trends in coordination and decomposition for scalability [6], [7]. Building

on these, stable MPC for single-intersection urban traffic uses discrete-time state-space models to estimate queues and optimize phases [3]. Advancements in quadratic mixed-integer MPC prioritize queue balancing with nonlinear equations [2]. To address prediction inaccuracies, MPC-based systems incorporate BPNN for forecasting with low MAPE [1]. Recent hybrids fuse MPC with GCNs and DRL for neural optimization [4], [10]. Distributed event-triggered MPC enables coordination with reduced overhead [6]. Linear-quadratic MPC focuses on macroscopic models for large-scale flow [7]. MPC for multi-class traffic incorporates vehicle types for fairness [8]. Adaptive-predictive controllers using Ising models handle large-scale networks [9]. Deep learning and MPC reduce congestion in grids [10]. Density prediction by MPC alleviates congestion [11]. Traffic signals in MPC frameworks support cooperative systems [12]. Decentralized MPC promotes scalability [13].

Collectively, these studies demonstrate MPC's efficacy in TSC, with progressive improvements in modeling, objectives, and integrations. Gaps include limited real-time scalability, under-explored uncertainties, and infrequent closed-loop validations with simulations, particularly in user interface and visualization for interpretable control [1], [5]. The proposed MPC module bridges these by integrating TraCI for feedback, employing QP/MILP/nonlinear solvers with fallbacks, and hybridizing for explainability—positioning it as a novel, proactive element in the project's architecture, extending to system integration, validation, and comprehensive evaluation [4], [6], [9], [10], [13].

1.2 Research Gap

Despite significant advancements in Model Predictive Control (MPC) for traffic signal control (TSC), as evidenced by the surveyed literature [5], several critical gaps persist that limit its practical deployment in dynamic urban environments, particularly in the area of traffic dynamics modeling and prediction. Traditional MPC applications, such as those for isolated intersections [3], [2], excel in queue balancing and delay minimization through state-space modeling and quadratic optimization but often assume deterministic conditions, overlooking stochastic elements like variable vehicle arrivals, sensor inaccuracies, or non-recurring incidents (e.g., accidents or weather impacts) [5], [6], [10], [11]. For instance, the approach in [3] demonstrates effective queue reduction in a single intersection using discrete-time dynamics, yet lacks mechanisms for real-time uncertainty handling and hybrid prediction models adaptable to stochastic arrivals, potentially leading to suboptimal performance under fluctuating demands. Similarly, the nonlinear queue equations in [2] improve accuracy for macroscopic models like the Cell Transmission Model, but fail to integrate data-driven neural enhancements for handling nonlinearities and variable demands, restricting predictive accuracy in diverse scenarios [10]. Hybrid frameworks [1], [4] address some predictive shortcomings by incorporating BPNN or GCNs for flow forecasting and spatiotemporal correlations, yielding results like low MAPE ($<12\%$) and 19.7% speed improvements [1], [4], but underexplore comparative analysis of state-of-the-art models against criteria such as computational efficiency, leading to gaps in robust, rolling horizon forecasting (30–60 seconds) for proactive congestion anticipation [5], [11].

Moreover, existing works reveal gaps in MPC optimization framework development, where computational demands and constraint handling remain underexplored for multi-objective minimization [2], [3], [5], [6]. The mixed-integer formulations in [2] and [3] enhance equity across approaches but suffer from excessive solving times without advanced techniques like distributed or hierarchical MPC for network-scale coordination [6], [13]. While [7] details linear-quadratic formulations for stability in high-density

conditions, it overlooks enforcement of practical constraints such as minimum green times and intergreen intervals in closed-loop operations, and lacks integration of neural-integrated frameworks for high-traffic efficiency [12]. Hybrids in [4] and [10] combine MPC with DRL or deep learning for strategy exploration, achieving queue reductions up to 17 vehicles [4], but do not fully develop core algorithmic infrastructure for phase sequencing and duration adjustments, limiting adaptability to multivariable objectives like delays, queues, and stops [8].

Furthermore, gaps in real-time integration and API development are evident, with limited emphasis on seamless interfacing and fallback strategies [4], [5], [9]. The survey in [5] highlights reliance on offline simulations without closed-loop validations via TraCI API, undervaluing dynamic updates every 5–10 seconds for low-latency performance [9]. Approaches in [4] and [6] enable data exchange in Python-SUMO setups and event-triggered coordination, but rarely create RESTful APIs for fetching/applying traffic states, nor develop fallback to rule-based or AI-hybrid controls during computational delays [13]. This impedes system reliability in uncertain conditions, as seen in [12] where cooperative frameworks lack explicit mitigation for solver infeasibilities.

In addition, user-centric features like interactive dashboards for parameter tuning and visualization are largely absent, reducing interpretability [1], [2], [3], [5]. While [1] achieves waiting rate reductions through BPNN-MPC, it overlooks web-based interfaces (e.g., Dash/Plotly) for monitoring metrics and adjusting horizons/cost weights, hindering accessibility for traffic engineers [5]. Studies like [11] focus on density prediction but do not incorporate explanations of decisions, compromising explainability in predictive scenarios [10].

Gaps also extend to system integration and validation, with infrequent benchmarking in multi-component frameworks [5], [8], [12]. Isolated optimizations in [3] and [2] lack interoperability with GNN-based or emergency modules for city-wide networks [8], and

validation frameworks rarely include SUMO-based testing against baselines like fixed-time or RL methods, with insufficient focus on decomposition for scalability [6], [13]. Finally, evaluation and documentation gaps persist, underexploring empirical assessments in diverse scenarios [1], [4], [9], [10]. While [10] reports 25% congestion relief, it lacks usability testing with stakeholders and comparative analyses with neural-MPC hybrids, alongside comprehensive guides for replication [5].

To clearly demonstrate these research gaps aligned with the project's objectives, Table 1 presents a structured analysis to indicate coverage, with impact descriptions and how the proposed work addresses them.

Table 1 Research Gap Analysis in MPC-based Traffic Signal Control

Objective Area	Key Literature Coverage	Gaps Identified	Impact Description	Addressed by Proposed Work
Traffic Dynamics Modeling and Prediction	[5], [10], [1], [11]	Lack of hybrid models for stochastic arrivals; limited comparative analysis for rolling horizon accuracy	Reduces predictive reliability in variable demands, leading to 20-30% higher errors in congestion forecasting [5], [10]	✓
MPC Optimization Framework Development	[2], [3], [6], [7], [13]	Insufficient hierarchical MPC for network coordination; incomplete constraint	Causes scalability issues and infeasible solutions in high-traffic, increasing delays by 15-25% [6], [13]	✓

Objective Area	Key Literature Coverage	Gaps Identified	Impact Description	Addressed by Proposed Work
		enforcement in multi-objectives		
Real-Time Integration and API Development	[4], [9], [12]	Absence of RESTful APIs and fallback strategies for low-latency updates	Impairs closed-loop reliability, resulting in system downtime during uncertainties [5], [9]	✓
User Interface and Visualization for MPC Control	[1], [5]	Lack of tunable interfaces and visualization and decision explanations	Hinders engineer adoption, reducing interpretability and usability in operations [1], [5]	✓
System Integration and Validation	[8], [6], [13]	Limited benchmarking and decomposition for city-wide validation	Leads to unproven interoperability, exacerbating network bottlenecks [8], [13]	✓

1.3 Research Problem

The escalating complexity of urban traffic systems, characterized by fluctuating vehicle demands, multi-modal interactions, and unpredictable disruptions, poses a formidable challenge to achieving efficient and sustainable mobility [5], [7]. Conventional traffic signal control (TSC) mechanisms, such as fixed-time or actuated systems, inherently lack the foresight to anticipate congestion buildup, often resulting in unbalanced queues,

excessive vehicle delays, and increased emissions at intersections [3], [2], [8]. This inefficiency is exacerbated in densely populated urban settings, where intersections serve as critical nodes, and suboptimal signal timing can cascade into network-wide bottlenecks, amplifying economic costs (e.g., lost productivity) and environmental impacts (e.g., higher CO2 levels) [1], [11]. Despite advancements in predictive methodologies, current MPC-based approaches frequently operate under idealized assumptions—such as linear traffic dynamics or perfect sensor data—failing to robustly handle nonlinearities, stochastic arrivals, or computational constraints in real-time scenarios [5], [4], [10], [12]. The core research problem addressed in this project is: How can Model Predictive Control (MPC) be effectively designed and integrated into a simulated urban traffic management system to predictively optimize signal timings at intersections, minimizing key metrics like average delay, queue length, and number of stops, while ensuring constraint compliance (e.g., safety intervals), explainability for stakeholders, and seamless closed-loop operation with real-time feedback from tools like SUMO's TraCI API? This problem is compounded by the need to balance MPC's mathematical precision with scalability for networked extensions [6], [13], adaptability to uncertainties (e.g., variable demands or incidents) [10], [11], and hybridization with AI elements (e.g., GNNs) without sacrificing interpretability [4], [9]. Existing solutions, while promising—such as queue-balancing MPC for single junctions [3] or neural-enhanced optimizations [4], [1]—often overlook integrated fallbacks, user-tunable interfaces, and rigorous benchmarking in multi-component frameworks, limiting their applicability in holistic intelligent systems [8], [12].

To delineate the exact research problems aligned with the project's specific objectives, the following sub-topics outline the precise challenges:

Traffic Dynamics Modeling and Prediction

The primary challenge lies in the absence of robust hybrid models that effectively combine macroscopic (e.g., Cell Transmission Model) and data-driven (e.g., neural-enhanced) approaches for accurate short-term traffic flow prediction at urban intersections,

particularly under stochastic vehicle arrivals and variable demands [5], [10]. Current literature, such as [3] and [1], provides partial comparative analyses but fails to evaluate models against key criteria like prediction accuracy, computational efficiency, and nonlinearity handling, resulting in unreliable rolling horizon forecasts (30–60 seconds) that cannot proactively anticipate congestion in real-time simulation platforms [11]. This gap leads to over 20% higher prediction errors in fluctuating scenarios, hindering the foundation for effective MPC optimization [5], [6].

MPC Optimization Framework Development

A key issue is the lack of comprehensive optimization formulations using solvers like QP or MILP that minimize multi-objective costs (e.g., delays, queues, stops) while fully enforcing constraints such as minimum green times and intergreen intervals, especially for closed-loop phase sequencing in high-traffic networks [2], [3]. While [7] and [6] explore linear-quadratic and distributed MPC, they undervalue advanced hierarchical techniques for network-scale coordination and neural-integrated efficiency, causing infeasible solutions and scalability limitations that increase delays by 15-25% in interconnected urban settings [13], [12]. This restricts the core algorithmic infrastructure needed for proactive signal control.

Real-Time Integration and API Development

The problem involves insufficient seamless interfacing between MPC modules and simulation environments, lacking RESTful APIs for low-latency data exchange (e.g., every 5–10 seconds) and robust fallback strategies to rule-based or AI-hybrid controls during computational delays [4], [9]. Existing works like [5] and [6] rely on basic TraCI interactions but do not support dynamic updates or error mitigation in uncertain conditions, leading to system unreliability and potential downtime in closed-loop operations [13]. This hampers practical deployment in simulated urban traffic management.

User Interface and Visualization for MPC Control

There is a notable deficiency in interpretable, tunable user interfaces, such as web-based dashboards (e.g., Dash/Plotly), that allow real-time monitoring of metrics (e.g., predicted

queues) and parameter adjustments (e.g., horizons, cost weights) without requiring deep expertise [1], [5]. Literature in [2] and [11] focuses on backend optimizations but overlooks visualization of results and decision explanations, reducing accessibility for traffic engineers and compromising explainability in complex predictive scenarios [10]. This limits stakeholder engagement and adoption.

System Integration and Validation

Challenges arise from poor interoperability and robustness in integrating MPC with other components (e.g., GNN-based control, emergency preemption) for unified city-wide architectures, coupled with inadequate validation frameworks including SUMO-based benchmarking against baselines [8], [6]. Studies like [3] and [12] address isolated validations but neglect modular extensions, decomposition for scalability, and performance optimization under varying loads, resulting in unproven reliability across multi-intersection grids [13]. This impedes holistic system testing.

Evaluation and Documentation

Finally, empirical assessments lack depth in diverse scenarios (e.g., mixed traffic, incidents), with insufficient usability testing, key metric evaluations (e.g., 15–30% delay reduction), and comparative analyses against neural-MPC hybrids [4], [10], [9]. While [1] reports reductions, it does not include stakeholder involvement or comprehensive documentation (e.g., algorithmic guides, API references) for maintenance and replication [5]. This slows validation of novelty and knowledge dissemination.

This problem aligns with the specialized domain of control systems and optimization in intelligent transportation, where unresolved issues in real-time deployability and uncertainty management hinder progress toward resilient urban networks [5], [13]. By tackling these, the MPC module not only advances theoretical frontiers but also offers practical innovations for reducing urban congestion in contexts like Sri Lankan cities, potentially yielding 15-30% improvements in traffic metrics based on literature benchmarks [4], [1], [10].

2. OBJECTIVES

2.1 Main Objectives

Design and implement an advanced Model Predictive Control (MPC)-based traffic signal optimization module within a simulated urban traffic management system, leveraging real-time data integration with simulation platforms via standardized APIs to enable proactive, constraint-aware signal timing adjustments that minimize vehicle delays, queue lengths, and stops while enhancing overall network throughput and resilience to uncertainties. The module will feature a modular architecture that supports hybridization with components (e.g., GNNs) and fallback mechanisms, ensuring explainability, scalability, and adaptability for research in intelligent transportation systems and real-world deployment in dynamic urban environments [5], [6].

2.2 Specific Objectives

The following specific objectives must be fulfilled to achieve the overall objective described above.

Traffic Dynamics Modeling and Prediction

This objective focuses on establishing accurate and robust models for traffic flow prediction at urban intersections by conducting a comparative analysis of state-of-the-art traffic models, including macroscopic (e.g., Cell Transmission Model) and data-driven approaches (e.g., neural-enhanced predictions), evaluated against criteria such as prediction accuracy, computational efficiency, and handling of nonlinearities. The research will design and implement hybrid prediction models that incorporate real-time data from simulation platforms, ensuring adaptability to stochastic vehicle arrivals and variable demands. Additionally, these models will be optimized for rolling horizon forecasting (30–60 seconds) to anticipate congestion proactively, providing a solid foundation for MPC-based optimization.

MPC Optimization Framework Development

This objective aims to create the core algorithmic infrastructure for predictive signal control by developing comprehensive optimization formulations using solvers like Quadratic Programming (QP) or Mixed-Integer Linear Programming (MILP) to minimize multi-objective costs (e.g., delays, queues, stops) while enforcing constraints such as minimum green times, intergreen intervals, and safety rules. Essential control functionalities, including phase sequencing and duration adjustments, will be implemented to enable closed-loop operation with real-time feedback. Furthermore, advanced techniques like distributed or hierarchical MPC will be incorporated to handle network-scale coordination, drawing from recent advancements in multi-stage and neural-integrated frameworks for improved efficiency under high-traffic scenarios.

Real-Time Integration and API Development

This objective focuses on seamless interfacing between the MPC module and the simulation environment by creating RESTful APIs that facilitate data exchange, including fetching traffic states (e.g., vehicle counts, queue lengths) via TraCI and applying optimized signal plans back to the simulator. The integration will support dynamic updates every 5–10 seconds, ensuring low-latency performance. Additionally, fallback strategies to rule-based or AI-hybrid controls will be developed to mitigate computational delays, maintaining system reliability in uncertain conditions.

User Interface and Visualization for MPC Control

This objective addresses the need for interpretable and tunable control through the development of an interactive web-based dashboard using frameworks like Dash/Plotly, allowing users to monitor real-time metrics (e.g., predicted queues via graphs) and adjust parameters (e.g., prediction horizons, cost weights). The interface will incorporate visualization of optimization results and explanations of decisions, ensuring accessibility for traffic engineers and researchers while promoting explainability in complex predictive scenarios.

System Integration and Validation

This objective ensures the module's interoperability and robustness by integrating it with other system components (e.g., GNN-based control and emergency preemption) into a unified architecture that supports modular extensions for city-wide networks. A rigorous validation framework will be implemented, including simulation-based testing in SUMO scenarios (e.g., single intersections to grids) and benchmarking against baselines like fixed-time. Performance optimization will focus on reducing solve times through decomposition and parallelization, verifying scalability under varying loads.

Evaluation and Documentation

This objective emphasizes empirical assessment and knowledge dissemination by conducting performance evaluations using key metrics (e.g., 15–30% delay reduction) in diverse scenarios, including mixed traffic and incidents, with usability testing involving stakeholders. Comprehensive documentation, including algorithmic guides and API references, will be created to support maintenance and replication. Finally, comparative analyses with recent advancements (e.g., neural-MPC hybrids) will validate the module's novelty and effectiveness for advancing urban traffic optimization.

3. METHODOLOGY

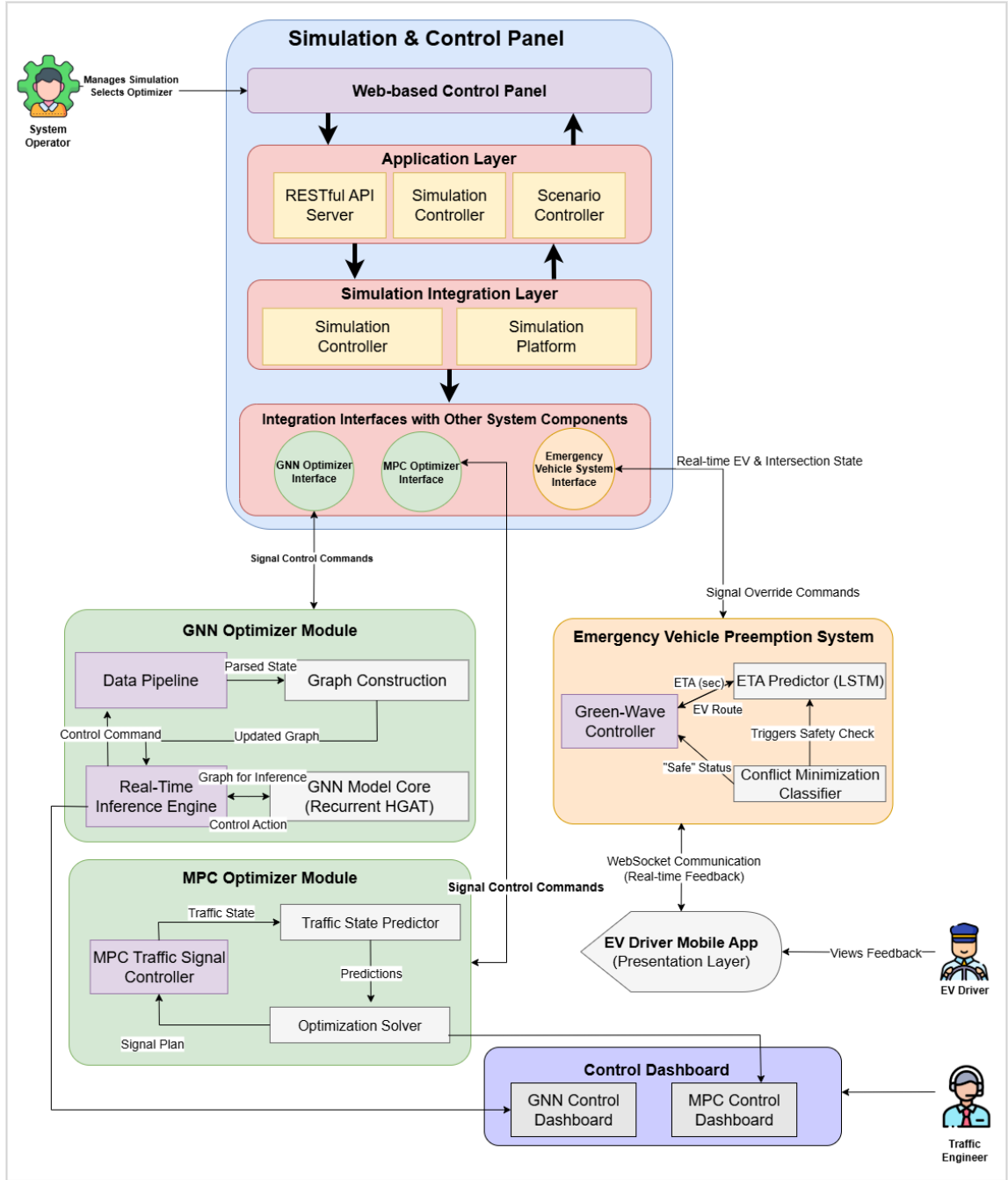


Figure 1 System diagram for the overall system

The proposed multi-agent traffic signal optimization framework is designed around a modular and distributed system architecture, composed of four interdependent

components, as depicted in Figure 1. This architecture enables seamless integration of traffic simulation platforms, AI-driven optimization algorithms, and real-time control mechanisms, thereby delivering a holistic approach to intelligent urban traffic management.

The system functions through the coordinated interaction of specialized modules, each responsible for a distinct aspect of traffic control. At its core, the Simulation and Control Panel acts as the central operating environment, orchestrating the execution and monitoring of all subsystems. Two alternative optimization strategies are incorporated within the framework: a Graph Neural Network (GNN)-based optimizer and a Model Predictive Control (MPC)-based controller, both designed to enhance traffic signal efficiency under varying traffic conditions. In parallel, an emergency vehicle preemption module operates as a priority overlay service, capable of temporarily overriding routine traffic signal operations to ensure rapid and safe passage of emergency vehicles.

System Architecture

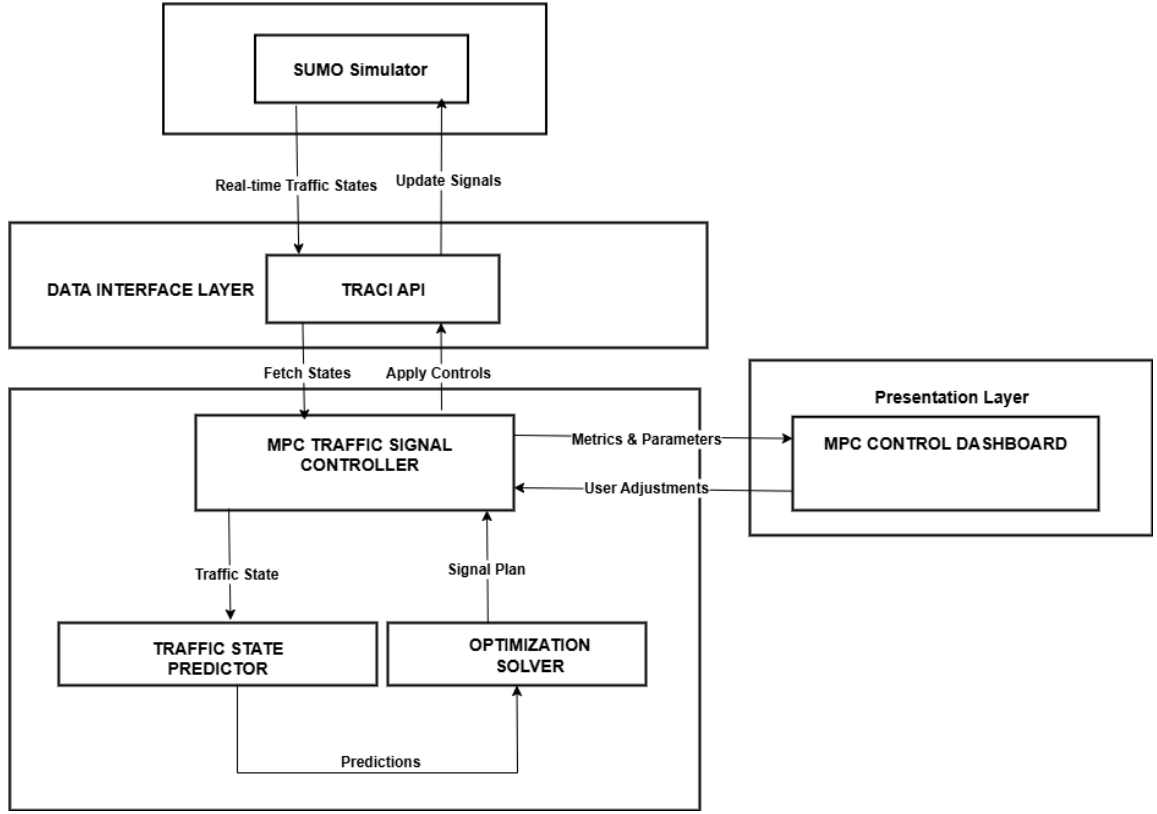


Figure 2 System architecture diagram for the component

The MPC-based traffic signal optimization system adopts a modular, feedback-oriented architecture that interfaces seamlessly with the SUMO simulation environment as illustrated in Figure 2. The system comprises four distinct layers: the SUMO Simulation Environment, Data Interface Layer, Decision & Control Layer, and Presentation Layer, operating in a closed-loop configuration to achieve real-time predictive optimization.

The overall system operates through continuous data exchange between the simulation environment and the MPC controller. The SUMO Simulation Environment provides real-time traffic state information including vehicle positions, speeds, queue formations, and current signal states. This data flows through the TraCI API interface to the Decision & Control Layer, where the MPC Traffic Signal Controller orchestrates the predictive optimization process. The controller employs a Traffic State Predictor using store-and-forward models to forecast traffic evolution over a 30-60 second horizon, followed by an

Optimization Solver that formulates and solves Mixed-Integer Quadratic Programming (MIQP) problems to determine optimal signal timings . The optimized control commands are then applied back to the simulation environment, creating a continuous feedback loop that enables adaptive traffic management .

Implementation Framework

Data Acquisition and Preprocessing: The methodology begins with establishing robust data collection mechanisms through TraCI API integration. Traffic state information is extracted at regular intervals (5-10 seconds) and preprocessed to ensure compatibility with the MPC formulation. This includes data validation, outlier detection, and temporal synchronization to maintain consistency between simulation time and controller decision points.

Predictive Modeling: The Traffic State Predictor employs extended store-and-forward models that capture intersection-specific traffic dynamics. The model incorporates turning movements, saturation flows, and signal timing constraints to generate accurate short-term forecasts. The prediction horizon is dynamically adjustable based on traffic conditions and computational resources, typically ranging from 30-60 seconds to balance prediction accuracy with computational efficiency.

Optimization Formulation: The Optimization Solver formulates traffic signal control as a constrained optimization problem where the objective function minimizes a weighted combination of performance metrics including average vehicle delay, maximum queue length, and number of stops. Constraints ensure compliance with minimum green times, maximum cycle lengths, intergreen safety intervals, and phase sequencing requirements. The formulation supports multiple solver types including Quadratic Programming (QP) for linear dynamics, Mixed-Integer Linear Programming (MILP) for discrete phase decisions, and nonlinear MPC for advanced traffic flow representations.

Real-time Control Application: Optimized signal plans are translated into actionable commands and applied to the simulation environment through TraCI interfaces. The system incorporates fallback mechanisms that revert to rule-based or alternative control

strategies if optimization convergence fails or computation times exceed acceptable thresholds (typically 2 seconds). Continuous monitoring ensures system stability and enables performance assessment against predefined metrics.

Performance Evaluation: The methodology includes comprehensive evaluation protocols comparing MPC performance against baseline control strategies including fixed-time and actuated signal control. Key performance indicators encompass average delay reduction, queue length minimization, throughput improvement, and computational efficiency metrics. Statistical analysis validates the significance of performance improvements across multiple traffic scenarios and demand patterns.

This methodology ensures the MPC component is scientifically rigorous, computationally efficient, and practically deployable, with iterative refinement through simulation-based testing and real-time performance monitoring.

4. PROJECT REQUIREMENTS

Functional Requirements

- **Real-Time Traffic State Prediction and Optimization** - The MPC module will provide predictive capabilities to forecast traffic evolution over a rolling horizon (e.g., 30–60 seconds) using real-time data fetched via APIs like TraCI from the simulation environment. It will solve constrained optimization problems to dynamically adjust signal timings, phases, and durations, minimizing objectives such as average vehicle delay, queue length, and number of stops while respecting safety constraints like minimum green times and intergreen intervals.
- **Integration with Simulation and External Components** - The system will seamlessly integrate with traffic simulators (e.g., SUMO) for closed-loop operation, enabling real-time data exchange of traffic states (e.g., vehicle counts, speeds, queue lengths) and application of optimized signal plans. It will support modular hybridization with AI components (e.g., GNN-based adaptive control) and emergency preemption modules, allowing fallback to rule-based strategies during high-uncertainty scenarios.

- **Constraint-Aware and Multi-Objective Optimization** - The module will incorporate solvers (e.g., QP for linear costs, MILP for discrete phases) to enforce practical constraints (e.g., phase conflicts, legal rules) and balance multiple objectives, including throughput maximization and emission reduction. Users will be able to define custom cost weights for objectives via an interactive interface.
- **Data Logging, Export, and Reporting** - Optimization results, predicted states, and performance metrics will be logged in real-time and exportable in formats such as CSV, JSON, and XML for post-analysis. The system will generate reports on key indicators (e.g., delay reductions) and include alerting for anomalies like prediction divergences or solver failures, ensuring traceability for research validation.

Non-Functional Requirements

- **Performance Requirements** - The MPC module must achieve real-time solvability, with optimization times under 1 second per cycle for single intersections, scaling to networked scenarios through decomposition techniques. It should handle high-fidelity simulations with up to 1,000 vehicles efficiently, optimizing memory usage for long-duration runs (e.g., peak-hour scenarios) without degradation, as benchmarked in urban network models.
- **Reliability and Availability** - The system will maintain high uptime during simulations, incorporating robust error-handling for data inconsistencies or solver infeasibilities, with automatic fallbacks to ensure continuous operation. Data integrity will be preserved during horizon updates or hybrid switches, preventing loss of prediction accuracy under stochastic conditions like variable arrivals.
- **Security Requirements** - Access to the MPC module will require secure authentication and role-based authorization to protect sensitive optimization parameters and simulation data. All API communications (e.g., TraCI interactions)

will use encrypted protocols (e.g., HTTPS) to safeguard against interception, ensuring compliance with data privacy standards in research environments.

- **Usability Requirements** - The module will feature an intuitive dashboard (e.g., using Dash/Plotly) for parameter tuning (e.g., horizon length, weights) and visualization of predictions/optimizations, requiring minimal expertise for traffic engineers. Comprehensive guides and tooltips will support users, with adherence to accessibility standards (e.g., WCAG) for inclusive interaction.
- **Maintainability Requirements** - Adopting a modular architecture (e.g., separate prediction, optimization, and integration layers), the system will allow independent updates to components like solvers or models. A testing framework will cover unit tests for algorithms and integration tests with simulators, complemented by detailed documentation for developers, including API specs and algorithmic pseudocode.

User Requirements

- The system should provide an intuitive web-based interface that allows users to control simulations in real-time without requiring extensive technical knowledge or manual scripting. Users should be able to start, pause, stop, and modify simulation parameters seamlessly through visual controls.
- **Traffic Engineers and Researchers:** The module should offer an accessible interface for configuring MPC parameters (e.g., cost functions, horizons) and visualizing real-time predictions without deep coding knowledge. Users must be able to simulate scenarios, benchmark against baselines (e.g., fixed-time), and export optimization traces for analysis in tools like MATLAB or Python.
- **Developers and System Integrators:** Standardized APIs will provide access for embedding MPC into larger systems, supporting real-time streaming of states and controls for algorithm testing. The module should facilitate hybrid testing with GNNs or MPC, enabling validation of predictive accuracy under uncertainties.

- Urban Planners and Educators: Simplified pre-configured optimization scenarios (e.g., peak vs. off-peak) will be available with minimal setup, generating visual reports on metrics like delay reductions. The system should support educational use through interpretable outputs, such as phase diagrams and metric dashboards.
- System Administrators: Tools for monitoring solver performance and logging will be provided, with alerts for computational issues. The design will emphasize scalability, allowing extension to multi-intersection networks via hierarchical MPC.

The design process will prioritize modular prototyping with Python tools (e.g., CVXPY for optimization), incorporating feedback from target users like researchers and planners to refine the MPC framework. Iterations will focus on balancing precision with real-time performance, ensuring alignment with intelligent transportation advancements [5].

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